

1.4 Basic Vector Calculus

Gradients, Jacobians, and Hessians in the notation this course uses.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

Notation follows Deisenroth, Faisal, and Ong, *Mathematics for Machine Learning*.

1. Gradient

The gradient is the derivative for several variables. Vary one coordinate of θ , hold the others. Those **partial derivatives**, stacked, are the gradient.

For $J(\theta): \mathbb{R}^l \rightarrow \mathbb{R}$,

$$\frac{\partial J}{\partial \theta_i}, \quad i = 1, \dots, l.$$

This course writes the collection as a **row**

$$\nabla J = \left[\frac{\partial J}{\partial \theta_1} \quad \frac{\partial J}{\partial \theta_2} \quad \dots \quad \frac{\partial J}{\partial \theta_l} \right] \in \mathbb{R}^{1 \times l}.$$

Some books write a column. Be consistent inside one derivation. Descent updates look like $\theta \leftarrow \theta - \alpha(\nabla J)^\top$ if ∇J is a row.

2. Chain rule

Let $J: \mathbb{R}^2 \rightarrow \mathbb{R}$ depend on $\theta_1(t)$ and $\theta_2(t)$. The derivative of J with respect to t is

$$\frac{dJ}{dt} = \left[\frac{\partial J}{\partial \theta_1} \quad \frac{\partial J}{\partial \theta_2} \right] \begin{bmatrix} \partial \theta_1 / \partial t \\ \partial \theta_2 / \partial t \end{bmatrix} = \frac{\partial J}{\partial \theta_1} \frac{d\theta_1}{dt} + \frac{\partial J}{\partial \theta_2} \frac{d\theta_2}{dt}.$$

That is a gradient times a Jacobian. Back-propagation (Module 6) is this rule applied to a long composition. You do not need the neural-net story yet. You do need to be willing to write the product.

3. Hessian

If $J(\theta_1, \theta_2)$ is twice differentiable, the **Hessian** is

$$\nabla^2 J = \begin{bmatrix} \partial^2 J / \partial \theta_1^2 & \partial^2 J / \partial \theta_1 \partial \theta_2 \\ \partial^2 J / \partial \theta_2 \partial \theta_1 & \partial^2 J / \partial \theta_2^2 \end{bmatrix}.$$

The mixed partials match (Schwarz), so the Hessian is **symmetric**. It measures **curvature** of J around a point.

At a critical point $\nabla J = 0$:

- $\nabla^2 J$ positive definite $\Rightarrow$ local **minimum**;
- negative definite $\Rightarrow$ local **maximum**;
- indefinite $\Rightarrow$ saddle.

Newton's method (Week 2) solves a linear system with this matrix every step. Eigenvalues of $\nabla^2 J$ are why a long thin valley makes steepest descent zigzag.

4. Practice

1. If a book prints ∇J as a column, how does the descent update $\theta \leftarrow \theta - \alpha \nabla J$ change compared with the row convention in these slides?
2. In the chain rule, which factor is "how J depends on θ " and which is "how θ depends on t "?
3. Why does a symmetric Hessian with one positive and one negative eigenvalue make a critical point a saddle, not a min?